

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1412

Kinsington 100 kV
April 28, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1412. The Interconnection Customer proposes a 46.2 MW battery energy storage facility interconnecting at the Kinsington 100 kV substation in the Illinois area. The request was received on February 24, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 505 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,573,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
RIV ROCK - RUBY STREET 100.0 kV line	JACOBS CORNER - KINSINGTON SUBSTATION 100.0 kV	125.1	150	100.0
JACOBS CORNER - KINSINGTON 100.0 kV L	KINSINGTON - RIVER ROCK 100.0 kV line	110.0	197	100.0
KINSINGTON - RIVER ROCK 100.0 kV	JACOBS CORNER - KINSINGTON 100.0 kV	110.0	197	100.0
ROCKPORT SATION - SAN LEANDRO U 100.0 kV line	BASALT - HOCKINSON 100.0 kV	108.0	133	19.8
PHELPS DODGE - EASY RD 230.0 kV	NRG ROCKFORD I - PHELPS DODGE 230.0 kV	104.5	625	12.9
ROCKPORT STA - MIDLOTHIAN SOUTH 100.0 kV	BASALT - HOCKINSON 100.0 kV Line	104.1	128	19.8
HOCKINSON - BASALT 100.0 kV L	ROCKPORT STA - SAN LEANDRO U 100.0 kV L	104.0	132	20.3
MIDLOTHIAN SOUTH - FOREMAN NORTH SUBSTATION 100.0 kV	BASALT - HOCKINSON 100.0 kV	101.7	123	19.8
NRG ROCKFORD I - LAS GALLINAS 230.0 kV	MAZOMNIE SUBSTATION - NRG ROCKFORD I 230.0 kV Line	101.4	631	5.2
ROSS AVENUE SUBSTATION - BASALT 100.0 kV	ROCKPORT STA SUBSTATION - SAN LEANDRO U 100.0 kV Line	100.5	128	20.3
FOREMAN N - RIO HONDO 100.0 kV Line	BASALT - HOCKINSON 100.0 kV	100.0	115	19.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RIV ROCK - RUBY STREET 100.0 kV	Rebuild line with higher-capacity conductor	\$3,321,000	19% (\$641,000)
JACOBS CORNER - KINSINGTON 100.0 kV L	Reconductor line	\$4,752,000	19% (\$918,000)
KINSINGTON - RIVER ROCK 100.0 kV	Reconductor line	\$1,515,000	19% (\$293,000)
ROCKPORT STA - SAN LEANDRO U 100.0 kV	Reconductor line	\$4,682,000	9% (\$425,000)
PHELPS DODGE - EASY RD 230.0 kV	Reconductor line	\$11,480,000	3% (\$341,000)
ROCKPORT STA - MIDLOTHIAN S SUBSTATION 100.0 kV	Reconductor line	\$2,361,000	9% (\$214,000)
HOCKINSON - BASALT 100.0 kV L	Reconductor line	\$4,040,000	8% (\$322,000)
MIDLOTHIAN SOUTH - FOREMAN NORTH 100.0 kV	Reconductor line	\$1,557,000	9% (\$141,000)
NRG ROCKFORD I - LAS GALLINAS 230.0 kV	Reconductor line	\$3,174,000	1% (\$28,000)
ROSS AVENUE SUBSTATION - BASALT 100.0 kV	Reconductor line	\$1,539,000	8% (\$123,000)
FOREMAN N - RIO HONDO 100.0 kV Line	Reconductor line	\$1,401,000	9% (\$127,000)
	Total allocated to project		\$3,573,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.